232 KNEC PHYSICS SYLLABUS

FORM 1

1. INTRODUCTION TO PHYSICS

1. physics as a science (reference to primary science syllabus)
2. Meaning of physics
3. Branches of physics
4. Relationship between physics and other subjects and technology
5. Career opportunities in physics
6. Basic laboratory safety rules

2. MEASUREMENTS

1. Definition of length, area ,volume, mass, density and time
2. SI units and symbols
3. Estimation of quantities
4. Conservation of units
5. Measuring instruments
6. Experiments on density
7. Problems on density

3. FORCES

1. Definition of force
2. Types of forces including cohesive .adhesive and surface tension
3. Experiments to demonstrate cohesion, adhesion and surface tension)
4. Effects of force
5. Mass ,weight and their relationship
6. Scalar and vector quantities
7. Problems involving W=mg (take g=10n/Kg)

4. PRESSURE

1. Definition of pressure
2. Pressure in solids
3. Factors affecting pressure in fluid
4. Derivation of P=gh
5. Atmospheric pressure
6. Simple mercury barometer
7. Application of pressure drinking straw syringe siphon, hydraulic press hydraulic brakes bicycles pump force, pump lift pump
8. Problems on pressure

5. PARTICULATE NATURE OF MATTER

1. Experiments to show that matter is made up of tin particles (eg cutting papers into small pieces dilution experiments
2. Brownian motion
3. Diffusion (Grahams law not required)

6. THERMAL EXPANSION

1. Temperature
2. thermometers
1. Liquids
2. clinical,
3. sixs maximum and minimum
3. Expansion of solids liquids and gases
4. Effects of expansion and contraction
5. Unusual expansion of water (anomalous expansion)
6. Application of thermal expansion include bimetallic strip

7. HEAT TRANSFER

1. Heat and temperature
2. Modes of heat transfer
3. Factors affecting heat transfer
4. Application of heat transfer on
1. Vacuum flask
2. Domestic hot water system
3. solar concentrators

8. RECTILINEAR PROPAGATION OF LIGHT AND REFLECTION AT PLANE

1. Rectilinear propagation of light

FORM 2

1. MAGNETISM

1. Magnets properties and uses
2. Magnetic and non magnetic materials
3. Basic law of magnetism
4. Magnetic field patterns
5. Magnetization and demagnetization
6. Domain theory of magnetism
7. Care of magnets
8. Constriction of a simple compass

2. MEASUREMENT II

1. Measurement of length using Vernier calipers and micrometer screw gauge
2. Decimal places ,significant figures and standard form
3. Estimation of the diameter of the molecule of oil (related to the size of the HIV virus mention effects of oil spill on health and environment
4. Problems in measurements

5. Project Work

1. construct Vernier calipers

3. TURNING EFFECT OF A FORCE

1. Moment of a force unit of moment of a force
2. Principle of moments
3. Problems on principle of moments (consider single pivot only)

4. EQUILIBRIUM AND CENTRE OF GRAVITY

1. Centre of gravity
2. State of equilibrium
3. Factors affecting stability
4. Application of stability
5. Problems on centre of gravity and moments of a force (consider single pivot only)

5. REFLECTION AT CURVED SURFACES

1. Concave and convex parabolic reflectors
2. Principal axis principal focus centre of curvature and related terms
3. Location of images formed by curved mirrors by
construction methods (Experimental on concave mirrors required)
4. Magnification formula
5. Application of curved reflectors

6. MAGNETIC EFFECT OF ELECTRIC CURRENT

1. Magnetic field due to a current
2. Oersteds experiment
3. Magnetic field patterns on straight conductors and solenoid(right hand grip rule)
4. Simple electromagnets
5. Factors affecting strength of an electromagnet
6. Motor effect (Fleming left hand rule)
7. Factors affecting force on a current carrying conductor in a magnetic field

8. Application :

1. Electric bell
2. Simple electric motors

9. Project work
1. construct an electromagnet and at least one of the following loudspeaker telephone receiver

7. HOOKES' LAW

1. Hooks law
2. Spring constant
3. Spring balance
4. Problems on Hooke's laws

8. WAVES I

1. Pulses and waves
2. Transverse and longitudinal waves
3. Amplitude (a), Wavelength ( ) frequency (f) periodic time (T)
4. V=f
5. problems involving v=f

9. SOUNDS

1. Sound nature and sources (Experimental treatment required)
2. Propagation of sound compressions and rarefactions
3. Speed of sound by echo method
4. Factors affecting speed of sound
5. Problems on velocity of sound

10. FLUID FLOW

1. Streamline and turbulent flow
2. Equation of continuity
3. Bernoulli's effect (Experimental treatment required)
4. Application of Bernoulli's effect Bunsen burner spray gun
5. Problems on equation of continuity

* FORM 3

1. LINEAR MOTION

1.Distance, displacement .speed velocity. acceleration (experimental treatment required)
2. Acceleration due to gravity
1. Free fall
2. Simple pendulum method
3. Motion time graphs:
1. Displacement time graphs
2. Velocity time graphs
4. Equations of uniform accelerated motion
5. Problems on uniform accelerated motion

2. REFRACTION OF LIGHT

1. Refraction of light laws of refraction (experimental treatment required)
2. Determination of refractive index
1. Snell law
2. Real apparent depth
3. Critical angle
3. Dispersion of white light
4. Total internal reflection and its effects -. Critical angle
5. Applications of total internal reflection

1. Prism periscope
2. Optical fibre

6. Problems on refractive index and critical angle

3. NEWTON'S LAWS OF MOTION

1. Newton's laws of motion
1. experimental treatment on inertia required
2. Conservation of linear momentum elastic collisions inelastic collisions
3. F=ma
4. Frictional forces:

1. Advantages and disadvantages,
2. Viscosity
3. Terminal velocity qualitative treatment)
5. Problems on Newton's laws and law of conservation of linear momentum (exclude problems on elastic collisions)

4. WORK ENERGY POWER AND MACHINES

1. Scale reading ammeter voltmeter
2. Electric circuits current potential difference
3. Ohms law (experimental treatment required)
4. Resistance types of resistors measurements of resistance and units
5. Electromotive force (emf)and internal resistance of a cell (E=V+Ir)
6. Resistors in series and in parallel
7. Galvanometers: Conversion to ammeters and Voltmeters
8. Problems on Ohms law resistors in series and in parallel

5. WAVES II

1.Properties of waves including sound waves :reflection refraction diffraction interference Experimental treatment required)
2. Constructive interference and destructive interference
3. Stationary waves (qualitative and experimental treatment required)

6. ELECTROSTATICS II

1. Electric field patterns
2. Charge distribution on conductors spherical and pear shaped conductors
3. Action at points lightning arrestors
4. Capacitance unit of capacitance (farad microfarad) factors affecting capacitance
5. Applications of capacitors
6. Problems on capacitors (using Q=CV, Ct=C1+C2)

7. HEATING EFFECT OF AN ELECTRIC CURRENT

1. Simple experiments on heating effect
2. Factors affecting electrical energy W=VIt, P=VI
3. Heating devices electric kettle, electric iron bulb filament electric heater
4. Problems on electrical energy and electrical power

8. QUANTITY OF HEAT

1. Heat capacity specific heat capacity units Experimental treatment required)
2. Latent heat of fusion, latent heat of treatment necessary)
3. Boiling and melting
4. Pressure cooker refrigerator
5. Problem on quantity of heat (Q=MC Q=MC

6. Project work
1. Construct a charcoal refrigerator cooler)

9. GAS LAWS

1. Boyles law Charles law pressure law, absolute zero
2. Kelvin scale of temperature
3. Gas laws and kinetic theory of gases
4. Problems on gas laws including PV/T constant)

* FORM 4

1. THIN LENSES

1. Types of lenses
2. Ray diagrams and terms used
3. Image formed

1. Ray construction
2. Characteristics,
3. Magnification.
4. Determination of focal length Experimental treatment required)

1. Estimation methods
2. Lens formula,
3. Lens-mirror method

5. Human eye, defects short sightedness only)
6. Optical devices:
1. Simple microscope
2. Compound microscope,
3. The camera
4. Problems involving the lens formula and the
magnification formula

7. Project
1. construct a telescope

2. UNIFORM CIRCULAR MOTION

1. The radian, angular displacement, angular velocity
2. Centripetal forces; F=mv2/r, F=mrw2
3. Application of uniform circular motion
4. Centrifuge, vertical horizontal circles banked tracks (calculation on banked tracks and conical pendulum not required)
5. Problems solving
1. (apply F=mv2/r, F=mrw2)

3. FLOATING AND SINKING

1.Archimedes' principle law of floating (experimental treatment)
2. Relative density
3. Application of Archimedes principle and relative density
4. Problems of Archimedes principle

5. Project work
1. Construct a hydrometer

4. ELECTROMAGNETIC SPECTRUM

1. Electromagnetic spectrum
2. Properties of electromagnetic waves
3. Detection of electromagnetic (e.m) radiations
4. Applications of (e.m.) radiations
5. Problems involving c=f

5. ELECTROMAGNETIC INDUCTION

1. Simple experiments to illustrate electromagnetic induction
2. Induction emf
1. Faradays law,
2. Lenz's law
3. Mutual induction
4. Alternating current generator, direct current generator
5. Flemming right hand rule
6. Transformers

7. application of electromagnetic induction
1. Induction coils
2. moving coil loudspeaker

8. **Problems on transformers**
9. Project work
1. Construction a simple transformer

6. MAINS ELECTRICITY

1. sources of mains electricity
2. Power transmission
3. Domestic wiring system
4. Kw-hr, consumption and cost of electrical energy
5. Problems in mains electricity
6. Excursion a field trip to a power station recommended

7. CATHODE RAYS AND CATHODE RAY TUBE

1. Production of cathode rays, cathodes ray tube
2. Properties of cathodes rays
3. C.R.O and TV tubes
4. uses of C.R.O
5. Problems on C.R.O

8. X-RAYS

1. Production of X rays ,X-ray tubes
2. Energy changes in an x-ray tube
3. Properties pf X-rays, soft X-rays and hard X-rays

4. Dangers of X-rays (Braggs law not required)
5. Problems on X-rays

9. PHOTOELECTRIC EFFECT

1. Photoelectric effect. protons, threshold frequency: work
2. Factors affecting photoelectric emission
3. Energy of photons
4. Einstein equation
1. hf=hf+1/2 mv2

5. Application of photoelectric effects
1. photo emissive
2. photo conductive
3. photovoltaic cell

6. Problems on photoelectric emission
7. Project work

10. RADIO ACTIVITY

1. Radioactive decay
2. Half life
3. Types of radiations, properties of radiations
4. Detectors of radiation
5. Nuclear fission ,nuclear fusion
6. Nuclear equations
7. Hazards of radioactivity, precautions
8. Applications
9. Problems on half life (integration nor required

11. ELECTRONICS

1. Conductors semi conductors, insulators
2. Intrinsic and extrinsic semi conductors
3. Doping
4. p-n junction diode
5. Applications of diodes half wave rectification and full wave rectification
6. Project work